

Web – Client/server interactions

May 6, 2026

1. Manual download of a web page using HTTP/1.1.

- a. The `telnet` command can be used to open a TCP connection towards a web server.

`telnet hostname port-number`

- b. The `openssl` command can be used to open a secure connection towards a web server.

`openssl s_client -connect hostname:port-number -crlf`

See the man pages for more details.

Check the traffic generated by the transactions using Wireshark and the content of web server logs. Analyze the headers sent in the HTTP request and response messages.

2. Download a web page using a browser (e.g., Mozilla Firefox) and HTTP/1.1.

- a. Remove all browsing data

`Settings -> Privacy & Security -> Clear browsing data`

- b. Change browser settings

`about:config`

Check the traffic generated by the transactions using Wireshark and the content of web server logs. Check the number of TCP connections opened by the browser. Analyze the headers sent in the HTTP request and response messages.

3. Download the home page of a “commercial” website using a browser and HTTP/1.1.

- a. Analysis of the HTTP requests and responses

`More tools -> Web Developer Tools`

- b. Display the browser cache

`about:cache`

How many and what type of objects are referenced by the HTML file? Are the objects stored on the same Web server? How many TCP connections does the browser open? Analyze the headers sent in the HTTP request and response messages. Are the objects cacheable? Analyze the content of the browser cache. Analyze the Page Load Time and study the effects of changing the number of parallel TCP connections.